

AI-Powered Credit Card

Recommendation Engine.

Product Requirement Document, User Stories & Prioritization.
Designed for FreechargeBiz, a subsidiary of Axis Bank.

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LIVE PROTOTYPE	https://visionary-dolphin-2dd706.netlify.app/cardmatch.html

◆ *The right card for how you actually spend.*

Executive summary

The problem. Choosing a credit card in India is a high-stakes decision wrapped in a broken experience. A 2025 SaveSage survey found that **~60% of Indians take whichever card their bank or employer offers**, only ~5% research before applying, and 40–50% use the same card for every transaction. The market is structurally underserved — almost nobody chooses optimally, and the few who try face a comparison journey that is generic, high-friction, and low-trust.

The product. CardMatch is an AI-powered credit card recommendation engine that reads a user's financial footprint — SMS, bureau, and Account Aggregator signals — and delivers a personalised, pre-eligible Axis Bank card recommendation in under a minute. The journey inverts the marketplace: lead with value, ask for the minimum data needed, and defer all hard friction until the user has decided to apply.

How it earns trust. Three architectural decisions separate this from a typical questionnaire-based engine:

1. **AI explains, scoring decides.** The AI builds the spend profile and writes the personalised “why this fits you” rationale. But the card itself is picked by a transparent, rule-based value calculation — not a model — making every recommendation auditable.
2. **Soft inquiry first.** Eligibility is checked before recommending, on a soft credit pull that doesn't affect the user's score. We never recommend a card the user would be rejected for — the structural fix to the wasted-inquiry problem.
3. **Account Aggregator consent.** Bank data is pulled through India's RBI-regulated AA framework: explicit, purpose-limited, time-bound, revocable consent. Raw SMS is parsed on-device. Account numbers are never stored.

Deliverables in this document

This Word document contains the Product Requirement Document, User Stories with Acceptance Criteria, and a Prioritization Matrix using MoSCoW and RICE. A fully working, hosted interactive prototype demonstrating the data-collection flow and the recommendation page is live at <https://visionary-dolphin-2dd706.netlify.app/cardmatch.html>.

01

Problem statement

Choosing a credit card in India is a high-stakes decision wrapped in a broken experience — and the market data shows just how broken. India crossed roughly **114.9 million credit cards in circulation by November 2025, with monthly spends hitting a record ₹2.17 trillion in September 2025**, yet almost nobody chooses their card well.

60% take the card their bank offers.

Only ~5% research before applying. 40–50% use the same card for every transaction.

Source: SaveSage India Credit Card Trends, April 2025.

That single finding reframes the problem: this isn't "forms are annoying." The market is **structurally underserved**. The overwhelming majority of cardholders never make an optimal choice, and the few who try face a comparison journey fractured in three ways:

- **Generic.** Comparison tables rank cards by popularity, not by the user's actual spending — so the "best card" is the same for everyone, even though the real drivers of Indian card value (online shopping, travel, food delivery) vary enormously by person.
- **High-friction.** Rigid questionnaires demand income, employment, and preferences upfront, before delivering any value — driving steep mid-funnel drop-off.
- **Low-trust.** Users apply for cards they're then rejected for (a wasted hard inquiry, a lost conversion), because eligibility was never checked before the recommendation.

The opportunity is to invert this journey: lead with value (an instant, personalised, pre-eligible recommendation), ask for the minimum data needed, and make the reasoning transparent. For a platform like FreechargeBiz operating in a market this large and this fast-growing, even small conversion improvements compound into significant volume.

02

User personas

These are assumption-based proto-personas, not research-backed. With no access to FreechargeBiz's primary user data, each is anchored to published segment data so design decisions rest on real signals rather than guesses.

Primary — “Aarav, the first-serious-card seeker”

26, software engineer in Bengaluru, ₹11L/year. Has one basic card from his salary account but suspects he’s leaving money on the table. Spends heavily online (Flipkart, Swiggy, Uber), takes 2–3 flights a year, and has no patience to compare 40 cards by hand.

Goal: the one card that pays him back the most for how he already spends.

Frustration: every site shows him the same five cards, and he doesn’t know which he’ll actually be approved for.

Why this persona. He sits at the centre of card growth and adoption. **~45% of new credit card holders in India are under 30**, many of them first-time borrowers (CRIF High Mark, 2024). Adoption concentrates in exactly his profile — metro, salaried — where banks market cards aggressively and eligibility is simpler. His online/food/travel spend mix is precisely what the engine reads to differentiate one card from another.

Secondary — “Priya, the cautious upgrader”

30, marketing manager, ₹15L/year, already holds a cashback card and is travel-curious. More deliberate — she wants to understand the fee-vs-perks trade-off before committing.

What wins her: a clear side-by-side of the recommended card against one alternative, with the value math shown.

Why this persona. She reflects where reward behaviour is heading. In 2025, **travel redemptions overtook cashback, with flights becoming India’s number-one redemption choice** — making a travel card like Axis Atlas a credible, aspirational next step for someone upgrading from a basic cashback card.

Out of primary scope — backlog persona

The thin-credit-file first-timer with no salary history. Deliberately excluded from the MVP so the recommendation logic and narrative stay sharp — calling out who we’re *not* building for is a conscious scoping decision.

03

Core features & user journey

Core features (MVP scope)

Feature	What it does
Consent-first onboarding	Granular, per-source consent (SMS, bureau, Account Aggregator) framed as the thing that enables personalisation, not a legal hurdle.
On-device SMS parsing	Raw SMS is categorised on the user's phone. Only the derived spend vector is transmitted — never message contents or account numbers.
Account Aggregator integration	Bank statement data pulled through India's RBI-regulated AA rails — explicit, purpose-limited, time-bound, revocable consent.
AI-driven profile builder	An LLM/NLP layer categorises messy merchant strings into a structured spend profile. Confidence-gated: when uncertain, the engine asks rather than guesses.
Soft-inquiry eligibility filter	Cards the user can't get are removed before ranking, using a soft credit pull that doesn't affect the user's score.
Value-based ranking	Each eligible card scored on estimated annual net value (rewards by category – fee after waiver). Transparent formula, not a black box.
Explainable recommendation	AI-generated "why this fits you" rationale grounded in the user's actual numbers and the scoring output. No hallucinated benefits.
Side-by-side comparison	Top pick presented alongside one strong alternative, with fees and value math both visible.
Deferred apply	PAN/KYC and the hard credit inquiry happen only when the user explicitly chooses to apply — never as a price of entry.

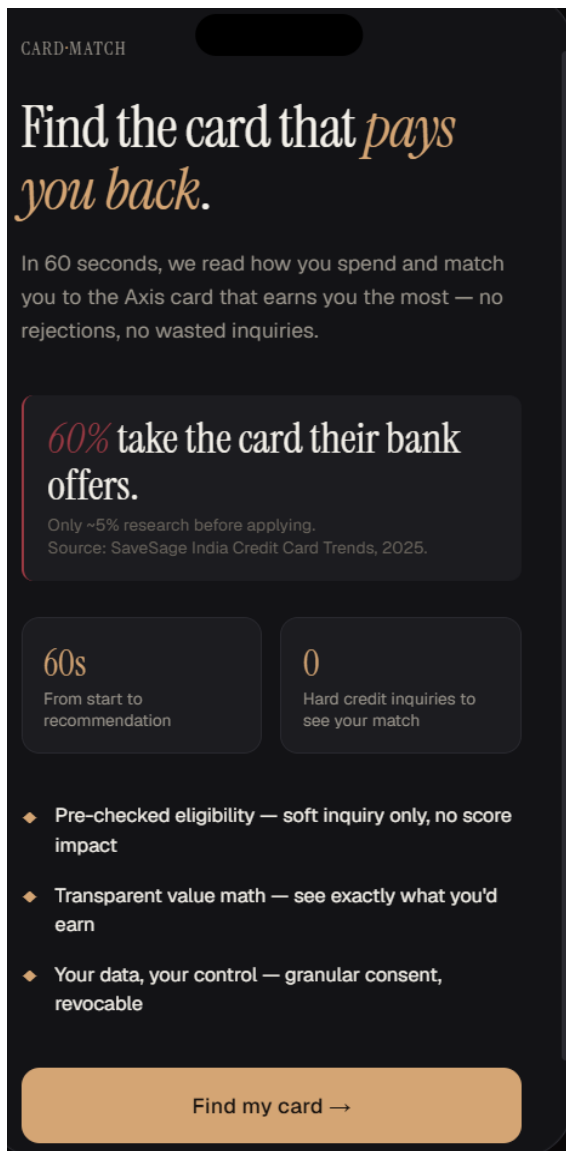
End-to-end user journey

The design spine is value before friction. Every screen either reduces effort or improves the recommendation. Every expensive data ask (PAN, full KYC) is deferred until after the user has seen the result and chosen to apply.

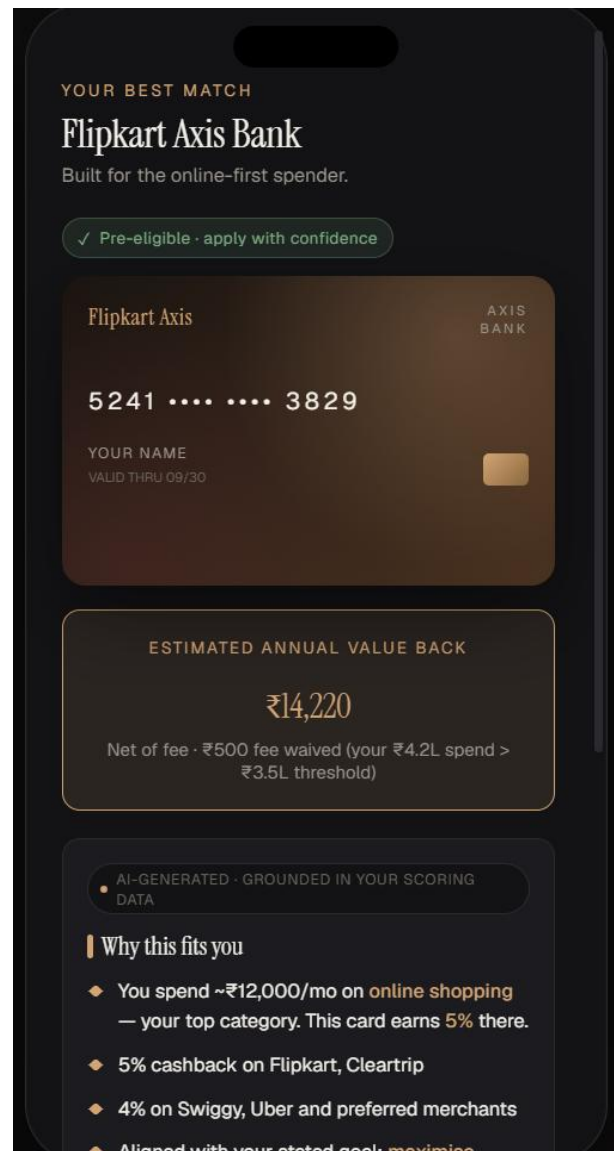
Step	What happens
1. Entry / hook	User lands from a FreechargeBiz prompt — “Find the card that actually pays you back, in 60 seconds.” Sets the expectation: fast, personalised, free. No personal field appears.
2. Consent, framed as benefit	A single transparent screen: connect SMS / bureau / bank data so we can read your pattern. Granular per-source toggles. Account Aggregator named explicitly. Skip-to-manual is always available.
3. Ingestion (the magic moment)	A brief “Reading your spending pattern...” animation as the engine simulates parsing SMS and bureau signals. The user typed nothing, yet we now know their top spend categories, monthly band, and eligibility.
4. Manual fallback / confirmation	Skippers go through a progressive one-question-at-a-time flow (top spend area, monthly band, primary goal). Auto-flow users see the inferred profile, with every category editable, before continuing.
5. Recommendation page	Hero card, plain-language “why this fits you” tied to actual data (“you spend ~₹X/month online — this card earns Y% there”), estimated annual value in rupees, eligibility badge, and one alternative for comparison.
6. Apply (deferred friction)	Only now do we collect PAN/KYC, and only because the user has chosen to proceed. The hard inquiry sits after value is delivered, with separate explicit consent.

Prototype: data-collection and recommendation in practice

The interactive prototype is live at <https://visionary-dolphin-2dd706.netlify.app/cardmatch.html>. Screens shown below are captured from the deployed build.



Landing — leads with the problem stat that justifies the product's existence.



Recommendation — personalised “why this fits you” with eligibility transparency.

04

Technical architecture & data strategy

This section answers the brief's explicit question — what data inputs feed the recommendation — and lays out the engine architecture, with a deliberate split between where AI earns its place and where deterministic logic is non-negotiable.

4.1 Data inputs

The engine recommends from four signal sources, layered from cheapest/least reliable to richest/most authoritative. The principle is progressive enrichment: a decent recommendation is possible from explicit input alone, and a great one becomes possible as the user grants access to deeper signals. The journey never blocks waiting for data.

Input	What it gives us	Used by	Reliability & privacy
Explicit user input <i>declared</i>	Top spend areas, monthly band, primary goal, existing card.	Profile builder, Matching.	High intent but recall-biased. Zero privacy risk.
SMS-derived signals <i>inferred</i>	Spend categories from merchant names, spend volume, salary-credit (income proxy), existing EMIs.	Profile builder.	Rich & passive. Parsed on-device ; raw SMS never leaves the phone.
Credit bureau data <i>consented</i>	Credit score, existing credit lines, utilisation, enquiry history.	Eligibility filter.	Authoritative. Pulled as a soft inquiry — no score impact.
Bank / transaction data <i>consented, via AA</i>	Actual spend by category, income inflows, EMI outflows.	Profile builder, Eligibility.	Most accurate spend picture. Shared via RBI-regulated AA consent rails.

4.2 The recommendation engine (five layers)

The load-bearing design decision — and the thing to defend in the pitch — is the **boundary between AI and deterministic logic**.

Layer	Type	What it does
1. Consent & ingestion	Input / rules	Captures explicit answers and, if opted in, kicks off SMS parsing and the AA / bureau pull. Consent is per-source and per-purpose.
2. Profile builder	AI	Turns raw signals into a structured spend profile: category distribution, monthly spend, income band. AI earns its place here — merchant strings (“AMETUR*SWIGGY BLR”) defeat rules but generalise under NLP. Guardrail: low-confidence categories trigger one clarifying question, not a guess.
3. Eligibility filter	Deterministic	Hard rule gate: minimum income, credit-score band, age, existing-card exclusions. Removes cards the user <i>cannot get</i> , so we never recommend a card they’d be rejected for. Rule-based on purpose — eligibility must be auditable, not probabilistic.
4. Matching & ranking	Deterministic	For each eligible card, compute estimated annual net value: <i>rewards earned across the user’s spend categories at that card’s rates, minus annual fee after waiver thresholds, plus goal-relevant perks</i> . Rank by net value, tie-broken by goal alignment.
5. AI reasoning	AI	An LLM generates the plain-language “why this fits you” narrative — but it does not pick the card . Layer 4’s scoring already chose the winner; the LLM only verbalises the numbers, grounded in that structured output. <i>The AI explains the recommendation; it never makes it.</i>

Why this AI / deterministic split matters

It gives a clean answer to the hardest reviewer question — “how do you know the AI won’t recommend a bad card?” **The AI never recommends. A transparent value calculation does.** AI is used where it’s genuinely better (understanding messy data; explaining results), and deliberately kept out of the layers that affect the user’s money.

4.3 Data strategy: consent, privacy, and trust

The whole product depends on users handing over sensitive financial data, so trust is the product. Four commitments:

Account Aggregator as the consent backbone. Rather than screen-scraping or asking users to upload statements, the engine pulls bank and transaction data through India's RBI-regulated Account Aggregator (AA) framework. The user (via an AA like Finvu or OneMoney) grants explicit, purpose-limited, time-bound, revocable consent. The bank (FIP) shares encrypted data with FreechargeBiz (the FIU). The AA itself is data-blind and only routes the consent and the encrypted payload. This is the compliant, user-controlled way to get the rich signal — and using it signals an understanding of the Indian fintech rails, not just generic “open banking.”

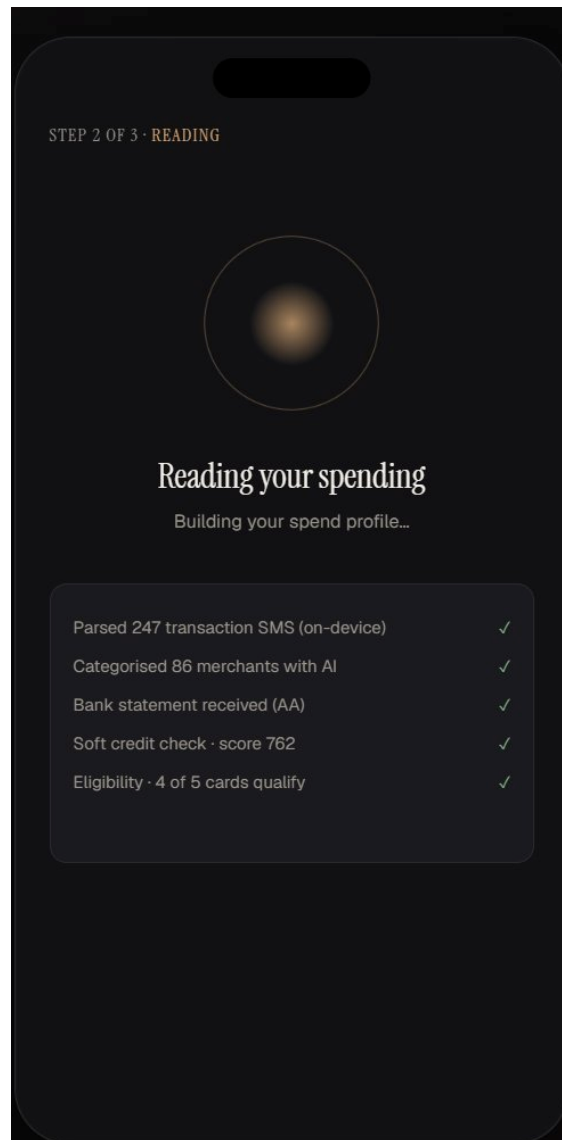
Soft pull now, hard pull only at apply. Recommendations run on a *soft* credit inquiry, which doesn't dent the user's score. The score-affecting *hard* inquiry happens only when the user actively chooses to apply. This is what structurally fixes the “low-trust / wasted inquiry” problem from the problem statement.

Data minimisation & on-device parsing. Raw SMS is parsed on the device; only the derived category vector (not message contents, never account numbers) is sent to the server. We store the derived profile (with consent) and never store credentials or full account numbers — which is precisely the promise the consent screen makes to the user.

AI governance. AI touches only the two layers where it's genuinely better (understanding messy data; explaining results) and is deliberately kept out of eligibility and ranking. Every AI output is either confidence-gated (Layer 2) or grounded in deterministic numbers (Layer 5), so the system stays explainable and auditable end-to-end — a non-negotiable for anything that influences a financial decision.

Prototype: the ingestion moment

The architecture above is rendered visibly in the prototype's ingestion screen: SMS parsed on-device, bank statement received via Account Aggregator, soft credit check (score 762), and the eligibility filter visibly reducing 5 candidate cards to 4 qualified ones — before the recommendation is ever computed.



Ingestion screen — the engine layers, visible to the user.

05

User stories & acceptance criteria

Organised by epic, with story IDs used in the prioritisation matrix that follows. Personas are referenced by role so every story traces back to a real user.

Epic A — Onboarding & consent

US-1 · Value before friction

As a first-time card seeker, I want to understand what I'll get before sharing anything, so that I feel safe enough to start.

Acceptance criteria

- Landing screen states the value proposition and a time estimate (~60 seconds) before any data is requested.
- No personal or financial field appears on the entry screen.
- A “how this works / privacy” link is reachable before consent.

US-2 · Granular, per-source consent

As a privacy-conscious user, I want to grant data access per-source and per-purpose, so that I stay in control of what I share.

Acceptance criteria

- Each source (SMS, bureau, bank-via-AA) has its own toggle with a plain-language purpose line.
- The user can proceed with zero sources connected (manual path stays open).
- Consent is shown as revocable and time-bound, with an explicit “we never store account numbers” statement.

Epic B — Data ingestion

US-3 · Automatic spend reading (the ‘magic’)

As a busy user, I want the app to read my spending automatically, so that I get a personalised result without filling forms.

Acceptance criteria

- On opt-in, an ingestion state shows progress and resolves within a few seconds.
- Raw SMS is processed on-device; only the derived category vector is transmitted.
- On completion, an inferred spend profile is presented for confirmation.

US-4 · Progressive manual fallback

As a user who skips connecting data, I want to answer a few quick questions one at a time, so that I'm not overwhelmed by a long form.

Acceptance criteria

- Questions are presented one per screen with a visible progress indicator.
- No more than five questions are needed to reach a recommendation.
- The user can go back and edit any answer without losing progress.

Epic C — Profile & engine

US-5 · Confidence-gated confirmation

As a user, I want to correct anything the engine misread about my spending, so that the recommendation is accurate.

Acceptance criteria

- The inferred profile is editable before the recommendation is generated.
- When category confidence falls below threshold, the engine asks one targeted clarifying question rather than guessing.

US-6 · Eligibility-first (soft inquiry)

As a user, I want to see only cards I can actually get, so that I don't waste a credit inquiry or face rejection.

Acceptance criteria

- The result set excludes cards failing income, credit-score, age, or exclusion rules.
- Eligibility runs on a soft inquiry only, with no score impact, and this is stated to the user.
- If no card qualifies, a graceful fallback (entry/secured card or guidance) is shown — never an empty screen.

US-7 · Value-based ranking

As a value-seeker, I want the recommendation ranked by what it actually pays me, so that it reflects my real spending.

Acceptance criteria

- Each eligible card is scored on estimated annual net value (rewards – fees after waivers + relevant perks) using the user's profile.
- The top card is selected by net value, with ties broken by the user's stated goal.
- The estimated annual value is shown as a concrete rupee figure.

Epic D — Recommendation & conversion

US-8 · Explainable recommendation

As a user, I want a clear reason why this card fits me, so that I trust it.

Acceptance criteria

- The “why it fits you” rationale references the user's actual data (e.g., “you spend ~₹X/month online”).
- Estimated annual reward value and net-of-fee figure are both displayed.

- The explanation is generated strictly from the scoring output — no benefit is shown that the card does not have.

US-9 · Side-by-side comparison

As a deliberate upgrader, I want to compare the top card against an alternative, so that I can weigh fees against perks before deciding.

Acceptance criteria

- At least one alternative is shown beside the top pick.
- The comparison shows fee, key reward rates, and estimated value for each.

US-10 · Deferred apply

As a ready user, I want to apply only after seeing my result, so that I share KYC/PAN only once I've decided.

Acceptance criteria

- KYC/PAN collection happens only after the recommendation, on an explicit “Apply” action.
- The hard inquiry is triggered only at this step, with separate explicit consent.

06

Prioritization

Using MoSCoW for the release cut (a crisp MVP-vs-backlog decision) and RICE on the non-obvious contenders (to show that sequencing isn't just gut feel). They work together: MoSCoW decides what ships; RICE helps order the maybes.

6.1 MoSCoW

Priority	Items	Rationale
Must have <i>MVP</i>	US-1, US-4, US-6, US-7, US-8	The smallest loop that proves the thesis: low-friction intake → eligibility-filtered → value-ranked → explained recommendation. Defensible without any “magic.”
Should have	US-2, US-3, US-5, US-9	High-impact enhancements — the SMS magic, granular consent, confidence-gating, and comparison — that elevate it from functional to differentiated.
Could have	US-10, AA bank-data integration, what-if spend sliders, save/share result	Valuable but deferrable; the apply handoff and full AA pull add real engineering and compliance load.
Won't have <i>this release</i>	Thin-file first-timer persona, vernacular languages, post-approval tracking, multi-card “wallet” recommendation	Explicitly out of scope to keep the MVP narrative sharp.

Reconciliation note for the demo

The prototype demonstrates the Must-haves *plus* US-3 (the magic ingestion), even though US-3 is a “Should.” This is deliberate: the data-collection screen is one of the two things being pitched, so the showpiece earns its place in the demo even though a strict product-MVP would ship the manual path first.

6.2 RICE (selected contenders)

Reach = estimated users per quarter reaching the feature. Impact on a 0.25–3 scale. Confidence as a percentage. Effort in person-weeks. Score = (Reach × Impact × Confidence) ÷ Effort. *Numbers are illustrative — the ordering is the point.*

Feature	Reach	Impact	Confidence	Effort	RICE score
US-9 Comparison view	90,000	2.0	95%	3	57,000
US-3 SMS “magic” ingestion	70,000	3.0	80%	8	21,000
US-5 Confidence-gated question	40,000	1.0	85%	2	17,000
US-10 Apply / KYC handoff	30,000	3.0	90%	5	16,200
AA bank-data integration	50,000	2.5	70%	13	6,731

Framework tension worth calling out

Pure RICE ranks the cheap, high-confidence comparison view above the SMS magic, because RICE rewards low effort. But the magic ingestion is a *strategic* inclusion — it’s the core differentiator and the friction-killer the entire product is premised on. MoSCoW therefore keeps it in scope, while RICE indicates that within a sprint, the comparison view should ship first. **Frameworks inform the order; product judgment decides what’s non-negotiable.**

07

References

Data points cited throughout this document. Note: these are secondary sources (finance publications aggregating RBI, CRIF High Mark, and survey data). Where possible, the underlying primary source is named in the reference.

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End of document. Live prototype: <https://visionary-dolphin-2dd706.netlify.app/cardmatch.html>